

PW 1 – Class Diagrams

Exercise 1: (Library)

A library needs to manage book loans by its patrons. Each book in the library has a title, author, publication date, and ISBN. Each patron of the library has a name, address, phone number, and a unique identification number. Each loan is associated with one book and one patron, and has a start date and a due date. A book can be loaned to zero or multiple patrons.

Exercise 2: (Transport company)

A transport company needs to manage its **trucks**, **drivers**, **customers**, and **orders**. Each truck of the company has a license plate number, a load capacity, and a status, which can be available or occupied. Each driver of the company has a name, address, phone number, driver's license number, and a hire date. Each customer of the company has a name, address, phone number, and a unique identification number. A transport order links a customer to a truck and a driver for a delivery on a specific date and address.

Exercise 3: (Event management)

An event management system needs to manage proposed **events**, user **registrations**, and **associated payments**. Each event has a name, date, venue, and cost. Users can register for specific events and make payments to confirm their participation. User registrations can be associated with different events.

Exercise 4: (Real estate agency)

A real estate agency needs to manage its properties, clients, and lease contracts. Each property has an address, property type, rental price, and living area. Each client has a name, address, phone number, and a unique identification number. A lease contract links a client to a property for a specific duration and cost.

Exercise 5: (Banking system)

A banking system needs to manage its customers' accounts, transactions, and credit cards. Each account has an account number, balance, and account holder. Transactions can be deposits or withdrawals on customers' accounts. Credit cards are linked to accounts and can be used to make transactions.

Exercise 6: (Hospital)

A hospital needs to manage its patients, doctors, nurses, appointments, medical records, treatments, medications, and departments. Each patient has a name, address, phone number, and a unique identification number. Each doctor and nurse has a name, ID, and specialization. Each appointment is a record of a patient visiting a nurse at a specific time. Each medical record contains information about a patient's medical history and treatments. Each treatment is a record of a patient receiving medical care for a specific condition (the doctor can provide treatment). Each medication is a record of a drug that is prescribed to a patient. The hospital has several departments, such as cardiology, pediatrics, and emergency.

Create a class diagram for the hospital system, including the relationships between the classes.

Exercise 7: (Company for transporting goods)

A company operates a fleet of vehicles that are used for transporting goods between warehouses and customers. The company needs to keep track of the vehicles, their maintenance schedules, the drivers who operate them, and the trips they make.

Each vehicle has a unique identifier (`vehicle_id`), as well as attributes such as make, model, year, and capacity. The company needs to know which vehicles are available, which are currently in use, and which are in need of maintenance or repairs.

Maintenance records are kept for each vehicle to track when routine maintenance tasks such as oil changes or tire rotations need to be performed, as well as when major repairs or inspections are due. The maintenance records include information such as the date of the maintenance task, the type of task, and the cost.

Drivers are assigned to operate the vehicles on specific trips. Each driver has a unique identifier (`driver_id`), as well as attributes such as name, address, and phone number. The company needs to keep track of which drivers are assigned to which vehicles on which trips.

Trips are recorded to track the movement of goods between warehouses and customers. Each trip includes information such as the starting location, destination, date and time of departure, date and time of arrival, and the driver and vehicle assigned to the trip. The company needs to be able to retrieve trip records to verify the delivery of goods and to track the performance of its fleet over time.